

# Sea of Okhotsk regional model validation

Pending alignment draft

| Field                              | Entry                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Region                             | Sea of Okhotsk, LME_052. The selected model describes the whole sea. <a href="#">Regional workbook</a>                                                                                                                                                                                                                                             |
| Catch source                       | Sea Around Us, 1950–2019 reconstructed catch. Reference: 2019 landings; all 151 exact reporting labels, including zero catch. <a href="#">Original catch</a>                                                                                                                                                                                       |
| Selected article                   | Chaikina (2004), A trophic mass-balance model of the Sea of Okhotsk for the 1980s period. BSc thesis, UBC, 49 pages. Original thesis unavailable. A related 2020 chapter documents the earlier model. <a href="#">Thesis listing</a>   <a href="#">Related chapter</a>                                                                             |
| Other known articles               | Radchenko (2015), Characterization of the Sea of Okhotsk Ecosystem Based on Ecosystem Modelling. A 48-group contemporary model is retained as an alternative; full reconstruction is unreviewed. <a href="#">Alternative article</a>                                                                                                               |
| Selected model                     | 52_1_Sea_of_Okhotsk_NE_(1980). NE means newly constructed detailed Ecopath. It is a whole-sea 1980s annual-average model with 29 source groups and computational import 30. Reported area: 1,590,000 km <sup>2</sup> . <a href="#">Accepted model</a>                                                                                              |
| Other models from the same article | The same source also compares a simplified nine-group SD 1980s model derived from the Shuntov–Dulepova food web. NE and SD are model structures, not northeastern and southern subregions. <a href="#">Simplified model</a>                                                                                                                        |
| Selection rationale                | Article — Retained as the broadest located whole-sea model despite incomplete original documentation.<br>Model — The detailed NE structure supplies finer biological resolution; existing numerical caveats remain.                                                                                                                                |
| Model extraction                   | All supplied native Table 1 B/PB/QB/EE values agree with accepted inputs. The accepted diet contains 74 normalized cells from eight printed columns totaling 0.998–1.001. Native catch fields and pollock cutoff are absent; export zeros, detritus B=1 and existing defaults/balance remain accepted assumptions. <a href="#">Source fidelity</a> |
| SPPR calculation                   | [MANUAL — Researcher: calculation choices and reasons]<br>[Excluded groups, scope and relevant values; other adjustments]                                                                                                                                                                                                                          |

| Field                       | Entry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GE diagnostics              | No negative SPPR entries across the basal-source columns. rho_living=0.86716588; b=0.15204353; Detritus 29 SPPR=1.25101369. <a href="#">Full source matrices</a>   <a href="#">Signed entries</a>                                                                                                                                                                                                                                                                                      |
| TE diagnostics              | Detritus 29 → Cottidae 14, Other Gadidae 12, Pinnipeds 4, Toothed whales 2; Phytoplankton 28 → Cottidae 14, Other Gadidae 12, Pinnipeds 4, Toothed whales 2; Phytobenthos 27 → Cottidae 14, Other Gadidae 12, Pinnipeds 4, Toothed whales 2. All 12 pairs are retained, including unfished mammals. TE FAIL remains. rho_living=1.02512005; b=0.00000000; Detritus 29 SPPR=0.94864761. b=0 is the TE convention. <a href="#">Full source matrices</a>   <a href="#">Signed entries</a> |
| Geographic fit              | A. Region covered by study: approximately 100% under the intended whole-sea proxy. B. Study covered by region: approximately 100% under that proxy. Author boundaries are unavailable; exact overlap is unmeasured and boundary confidence is Low.                                                                                                                                                                                                                                     |
| Temporal fit                | Static 1980s coefficients and fixed biomass proportions transfer across 1950–2019 whole-LME landings, catch and discards. Reference 2019 is roughly three decades later. Seasonal source averaging does not reconstruct annual catch composition or ecosystem change.                                                                                                                                                                                                                  |
| Other                       | GE and With Egestion return WARN; TE returns FAIL. All 120 source-recipient entries per method were inspected, including unfished groups and import. Inherited near-singular MC TE and negative/unsupported methods remain. Expanded mapping cannot approve these failed results. <a href="#">Method exposure and signed contributions</a>                                                                                                                                             |
| Open issues and next action | [Researcher: open issues and next actions]                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Review and reproducibility  | Researcher name: [name]   review date: [date]                                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Taxon mapping and coverage

Reference: 2019 landings; all source scope, all reporting labels, no group filter.

29 source groups (+1 computational import); 151 catch taxa.

Simple-chain PPR totals 248,368,608.38 t C in 2019.

20 labels lack a classic coefficient/TL; all have zero 2019 landings and zero annual contribution. Unknown coefficients and TL remain ?.

Method: protected independent simple trophic-chain coefficients.

[All 151 taxon mappings](#) and [descriptive Sources sheet](#)

| Overall confidence | Taxa | Catch percentage | PPR percentage |
|--------------------|------|------------------|----------------|
| High               | 46   | 33.0296%         | 64.0466%       |
| Medium             | 56   | 55.8556%         | 31.4082%       |
| Low                | 2    | 0.0901%          | 0.3575%        |
| Very low           | 46   | 11.0196%         | 4.1872%        |
| Unresolved         | 1    | 0.0051%          | 0.0005%        |

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

### Group assignment rules

| Plain-language rule                                                                | Confidence | PPR percentage |
|------------------------------------------------------------------------------------|------------|----------------|
| Literal member or verified common-name crosswalk to a named taxonomic source pool. | High       | 92.5385%       |
| Broad reporting union approximated across compatible source pools.                 | Very low   | 2.7698%        |
| Inferred eligible regional family, depth, habitat or feeding set.                  | Medium     | 1.8253%        |
| Represented ecological or lineage analogue with substantial mismatch.              | Very low   | 1.4174%        |
| Supported ecological extension to a represented feeding or habitat guild.          | Medium     | 0.9010%        |
| Partial or conflicting habitat fit with a material source mismatch.                | Low        | 0.3575%        |
| Supported extension from a named, closely related regional representative.         | Medium     | 0.1900%        |
| No defensible source membership or complete analogue set established.              | Unresolved | 0.0005%        |

Membership evidence: [native Table 1 and diet](#), [primary taxonomy and regional context](#) and [all candidate decisions](#)

## Allocation weight rules

| Plain-language rule                                                                                       | Confidence | PPR percentage |
|-----------------------------------------------------------------------------------------------------------|------------|----------------|
| One reviewed group receives all catch; no numerical split.                                                | High       | 67.4159%       |
| Complete source biomass after unavailable native catch; fixed composition and equal catchability assumed. | Medium     | 32.5836%       |
| No numerical allocation without established membership.                                                   | Unresolved | 0.0005%        |

Weights and assumptions: native group catches are unavailable. Canonical export=0 is not an observed stage fraction. Complete source biomass supplies every multi-group vector at Medium allocation confidence. Pollock supplies 48.9157% of 2019 landings and is allocated 53.874619% adult/nonjuvenile and 46.125381% juvenile, with equal catchability assumed and no recovered cutoff. Fixed 1980s proportions apply across all years and catch bases. Each rule table partitions the same simple-chain PPR. [Complete candidate vectors](#)

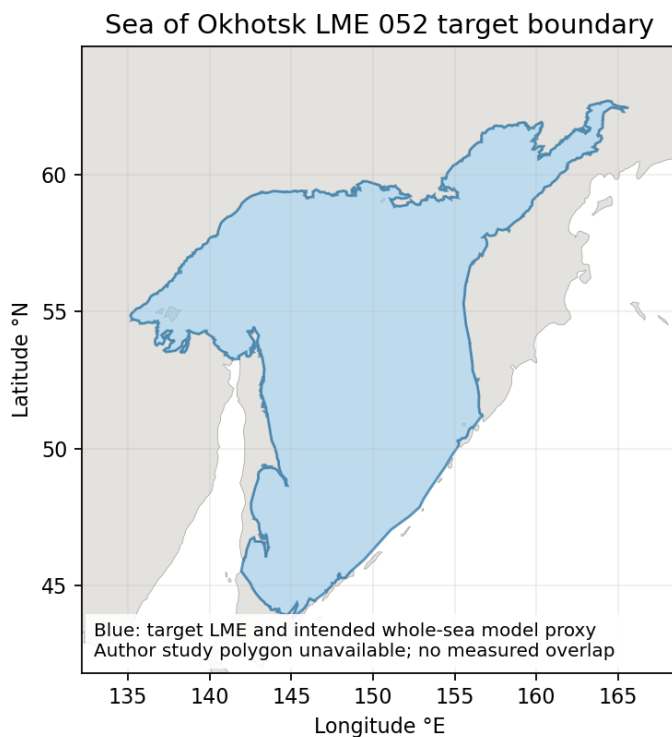
Unresolved: Cyprinidae, because unobserved freshwater family composition supplies no defensible local marine home. It retains 177.054315 t landings and 1,241.263560 t C classic PPR in the denominator. Very low: 46 labels, 11.0196% landings and 4.1872% classic PPR. These include broad bony/invertebrate unions and unrepresented cartilage, pelagic and grazing-fish analogues. Source Salmon is mostly zooplanktivorous, limiting tuna/shark transfer. Failed TE has signed caught contributions; ordinary positive confidence percentages are unavailable. [Flagged appendix rows](#) | [Method contributors](#)

# Geographic evidence screenshots

R is the Sea Around Us Sea of Okhotsk LME. S is described by the source as the whole sea. The displayed polygon reconstructs that intended domain using R; author GIS and a study-area map are unavailable. Approximate A/B values therefore describe the intended proxy, not measured overlap.

## Region area

### Target LME 052 and intended whole sea proxy



WGS84 target area: 1,563,743 km<sup>2</sup>. Source reported area: 1,590,000 km<sup>2</sup>. Their area ratio is 98.3486% (target/source); it is not an overlap calculation. No geographic multiplier is applied. [Target polygon](#) | [Qualified geographic assessment](#)

## Study area in the article

### Study area map unavailable native methods excerpt

#### Materials and Methods

The Okhotsk Sea is located on the north-eastern the of Asian continent. It is bound by the Russian Federation and Japan. According to Shuntov (2001), the Okhotsk Sea is located in the temperate climate zone. The total area of the region is estimated to be about 1,590,000 km<sup>2</sup>. The Okhotsk Sea is the coldest of all of the Russian Far East seas with more than 50% of the region covered by ice for almost one third of the year. The average depth of the sea is about 821 meters with maximum depth of 3916 meters (Dobrovolsky and Zalogin 1982).

Large phytoplankton blooms and active immigration of many species of fish and mammals occur in the region in the summer season. The Okhotsk shelf is one of the most productive regions in the Russian Far East. Because of the high productivity of the region, native people are highly dependent on the Okhotsk Sea resources (Okey 2003). Agriculture in the surrounding regions is not developed well enough to have a significant impact on the health of the ecosystem. Commercial fisheries have a considerable influence on the area. In the 1960s, total annual catch in the region was about 1 million t, increasing to 2.4-2.6 million t in 1980s. Some of the parts of the Okhotsk Sea, such as West Kamchatka suffer from especially intensive fishing effort. Catches ranged from 8 to 22 t·km<sup>-2</sup>·year<sup>-1</sup> in 1970s (Lapko and Radchenko 2000).

The 1980s were chosen because, due to the increased amount of funding available from the Soviet government, there was a vast amount of research carried out at that time. Also, the Okhotsk Sea did not appear to have suffered significant disturbances or changes during this period (Shuntov and Dulepova 1997).

Related 2020 chapter PDF25 / printed24. The methods describe the whole sea and report 1,590,000 km<sup>2</sup>. Chapter figures are food-web and simulation/results diagrams, not study-area maps. The excerpt supplies native geographic evidence while preserving the missing-map limitation. [Native chapter](#) | [Source identity review](#)

NE denotes the newly constructed detailed Ecopath model. The accepted identifier is retained; no northeastern boundary or original-thesis map is inferred.